

Caringbah High School

2016

Trial HSC Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen (Black pen is preferred)
- Board-approved calculators may be used
- A Board-approved reference sheet is provided for this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6 – 14

90 marks

- Attempt Questions 11–16
- Allow about 2 hour and 45 minutes for this section

Question 1 - 10 (1 mark each) Answer on the page provided.

1 If $a = 1 - 2c$, which expression has c been correctly made the subject?

(A) $c = \frac{-1 - a}{2}$

(B) $c = \frac{1 - a}{2}$

(C) $c = \frac{a - 1}{2}$

(D) $c = \frac{a + 1}{2}$

2 If $f(x) = 2x^2 - 3x + 4$, what is the value of $f(1) - f(-1)$?

(A) -6

(B) -2

(C) 6

(D) 2

3 The number of worms W in a worm farm, at time t is given by $W = 3000e^{-kt}$, where k is a positive constant.

Over time, which expression describes the change in the number of worms?

(A) decreasing at a constant rate

(B) increasing at a constant rate

(C) decreasing exponentially

(D) increasing exponentially

4 The second term and the fifth term of a geometric sequence are -3 and 192 respectively. What is the common ratio of the sequence?

(A) 4

(B) -4

(C) 8

(D) -8

- 5 Which of the following circles has the lines $x = 1$, $x = 5$, $y = 3$, and $y = 7$ as its tangents?

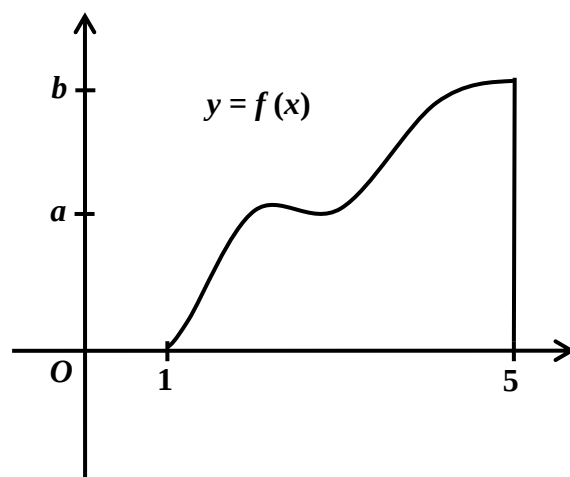
(A) $(x - 3)^2 + \left(y - \frac{7}{2}\right)^2 = 4$

(B) $\left(x - \frac{5}{2}\right)^2 + (y - 5)^2 = 4$

(C) $(x - 5)^2 + (y - 3)^2 = 4$

(D) $(x - 3)^2 + (y - 5)^2 = 4$

6



Using Simpson's rule with 3 function values, which expression best represents the area bounded by the curve $y = f(x)$, the x-axis and the lines $x = 1$ and $x = 5$?

(A) $\frac{2}{3}(1 + 4a + b)$

(B) $\frac{1}{2}(1 + 4a + b)$

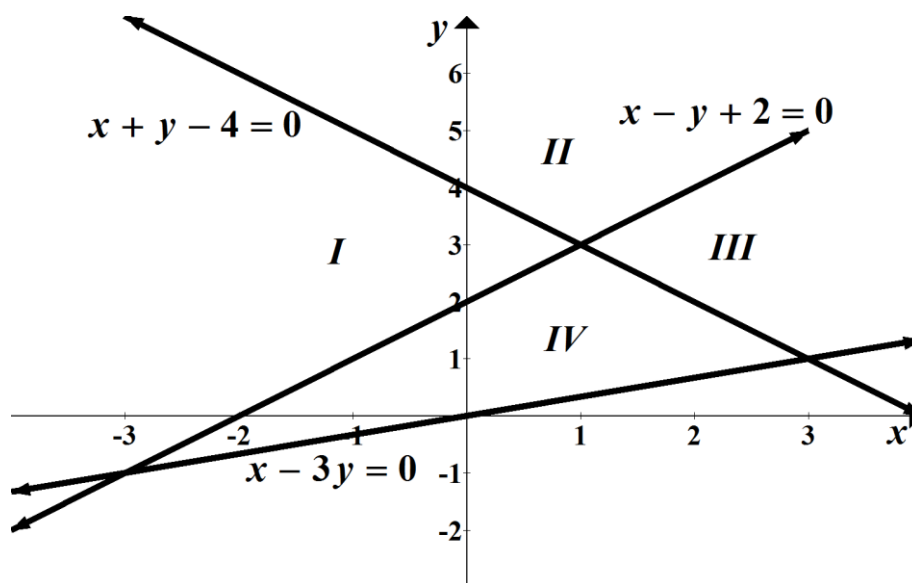
(C) $\frac{1}{2}(4a + b)$

(D) $\frac{2}{3}(4a + b)$

- 7 How many points of intersection of the graph of $y = \sin x$ and $y = 1 + \cos x$ lie between 0 and 2π ?

(A) 1 (B) 2
(C) 3 (D) 4

8

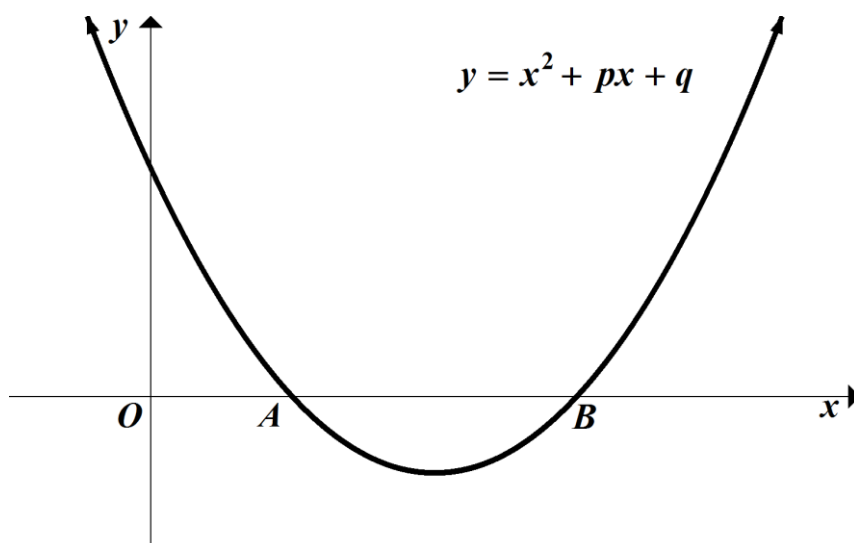


In the figure above, which region represents the solution to the following inequalities?

$$\left. \begin{array}{l} x - 3y < 0 \\ x - y + 2 > 0 \\ x + y - 4 > 0 \end{array} \right\}$$

(A) I (B) II
(C) III (D) IV

9



In the figure above, the graph of $y = x^2 + px + q$ cuts the x-axis at A and B.

Which is the value of $OA + OB$?

(A) $-p$

(B) p

(C) $-q$

(D) q

- 10 If $\log_{10} x, \log_{10} y, \log_{10} z$ form an arithmetic progression, which one of the following relationships hold true?

(A) $y = 10^{\frac{x+z}{2}}$

(B) $y = \frac{x+z}{2}$

(C) $y^2 = x + z$

(D) $y^2 = xz$

END OF MULTIPLE CHOICE QUESTIONS

Section II**90 marks****Attempt all questions 11–16****Allow about 2 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.**Marks**

- (a) Find the value of $e^{-\frac{\pi+3}{2}}$, correct to 3 significant figures. 2
- (b) Factorise $50 - 2x^2$. 2
- (c) Simplify $\frac{1}{x^2-1} - \frac{1}{x-1}$, expressing the answer as a single fraction. 2
- (d) Solve $|3x-5| = 4$. 2
- (e) Find the values of a and b if $\frac{1}{3-\sqrt{2}} = a + b\sqrt{2}$. 2
- (f) State the domain and range of $y = \sqrt{2x-1}$. 2
- (g) Find $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{3} dx$. 3

Question 12 (15 marks) Start a NEW booklet.**Marks**

(a) Find $\frac{d}{dx}(\sqrt{x^3 + 1})$. 2

(b) Find the limiting sum of $81 - 27 + 9 - \dots$. 2

(c) Find $\int (3x - 2)^5 dx$. 2

(d) Find the exact value of $\sin \frac{3\pi}{4} + \cos \frac{7\pi}{6}$. 2

(e) The gradient function of a curve is given by $3x^2 - 5$.
If the curve passes through the point $(1, 1)$, find its equation. 2

(f) When a balloon is being filled with helium, its volume at time t is given by

$$V = \frac{\pi t^3}{12} \text{ cm}^3, \text{ where } t \text{ is in seconds.}$$

Find the rate at which the balloon is being filled when $t = 1$. 2

(g) For the parabola $6y = x^2 + 2x + 13$, use completing the square or otherwise to find:

(i) the vertex. 2

(ii) the focal length. 1

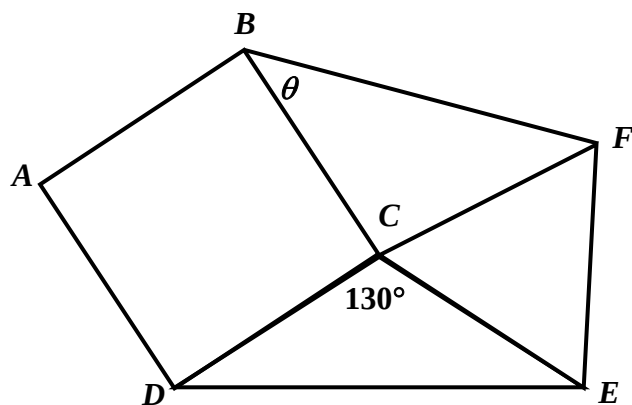
Question 13 (15 marks) Start a NEW booklet.

Marks

(a) Find $\frac{d}{dx}(e^{\cos x})$. 1

(b) Find the shortest distance of the point $(1, -3)$ from the line $2x - 5y = 4$. 2

- (c) In the figure below, $ABCD$ is a square, CEF is an equilateral triangle, $\angle DCE = 130^\circ$ and $DC = CE$.



Find the size of $\angle CBF$, giving clear reasons. 3

- (d) Find the value of k for which the equation $3x^2 + 10x + k = 0$ has:

(i) one root which is the reciprocal of the other. 1

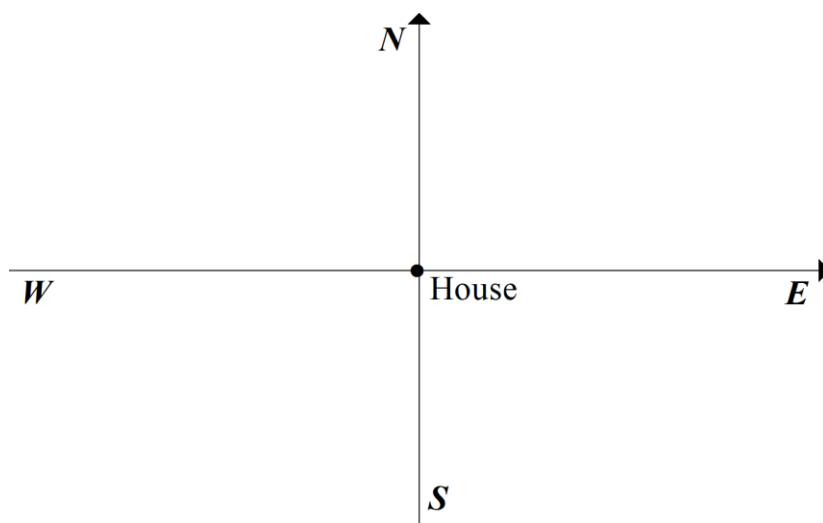
(ii) equal roots. 2

Question 13 continues on page 9

Question 13 (continued)

Marks

(e)



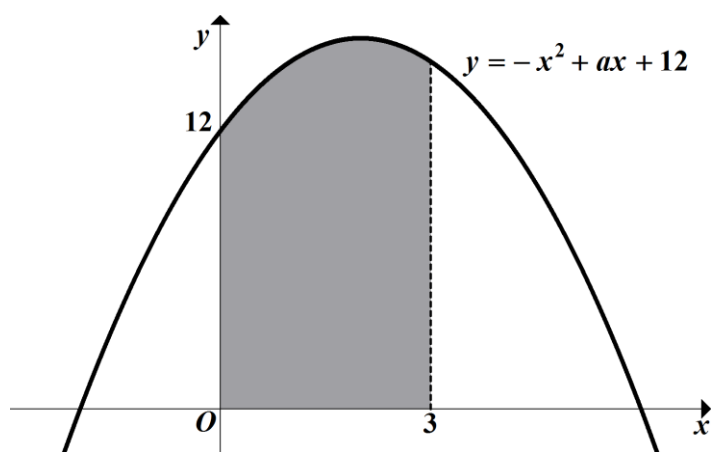
Maeve walks from her house for 6 km, on a bearing of 310° to point B .

She then walks on a bearing of 215° to a point C which is due west of her house.

(i) Copy the diagram and clearly indicate the information mentioned above. 1

(ii) Calculate to 1 decimal place the distance of C to her house. 2

(f) Part of the graph of the function $y = -x^2 + ax + 12$, is shown below.



If the shaded area is 45 square units, find the values of a .

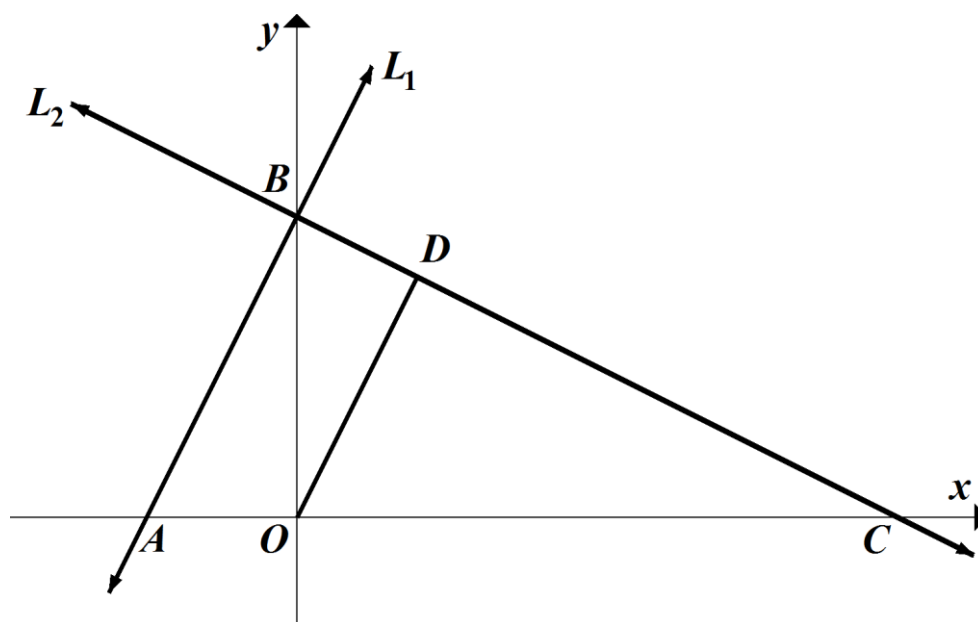
3

End of Question 13

Question 14 (15 marks) Start a NEW booklet.**Marks**

- (a) In the diagram below, the straight line L_1 has equation $2x - y + 4 = 0$ and cuts the x -axis and y -axis at A and B respectively.

The straight line L_2 , passing through B and perpendicular to L_1 cuts the x -axis at C . From the origin O , a straight line perpendicular to L_2 is drawn to meet L_2 at D .



- | | | |
|-------|---|---|
| (i) | Write down the coordinates of A and B . | 1 |
| (ii) | Show that the equation of L_2 is $x + 2y - 8 = 0$. | 2 |
| (iii) | Find the equation of the line OD . | 1 |
| (iv) | Show that the coordinates of D are $\left(\frac{8}{5}, \frac{16}{5}\right)$. | 1 |
| (v) | Hence or otherwise, find the area of quadrilateral $OABD$. | 2 |

Question 14 continues on page 11

Question 14 (continued)

Marks

(b) (i) State the period for $y = 3\cos\frac{x}{2}$. 1

(ii) Neatly sketch the graph of $y = 3\cos\frac{x}{2}$ for $0 \leq x \leq 2\pi$. 2

(c) Prove that $\frac{\tan \theta}{\sin \theta} - \cos \theta = \tan \theta \sin \theta$. 3

(d) Michael is attempting to solve the equation $2\ln x = \ln(3x + 10)$.

He sets his work out using the following correct steps:

- $\ln x^2 = \ln(3x + 10)$
- $x^2 = 3x + 10$
- $x^2 - 3x - 10 = 0$

He is now unsure of what to do next.

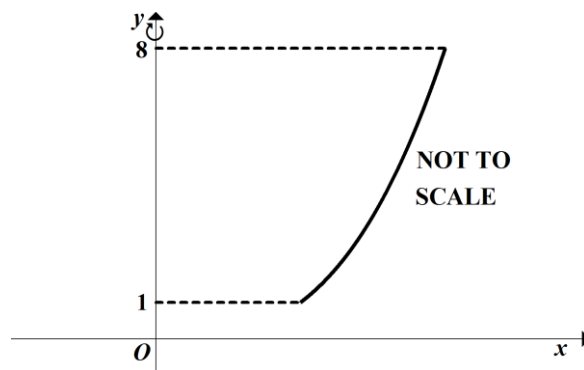
Complete the solution for Michael.

2

End of Question 14

Question 15 (15 marks) Start a NEW booklet.**Marks**

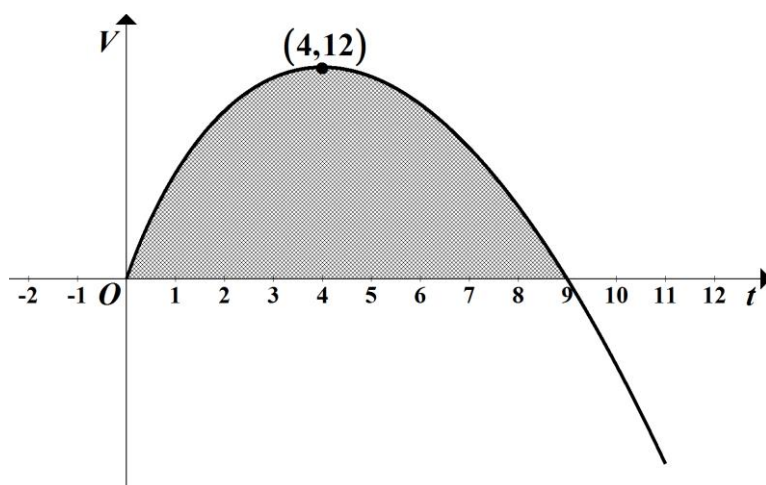
- (a) A solid is formed by rotating the curve $y = x^3$ about the y -axis for $1 \leq y \leq 8$.



Find the volume of the solid in exact form.

3

- (b)



A particle is observed as it moves in a straight line between $t=0$ and $t=11$.

Its velocity V m/s at time t is shown on the graph above.

- | | | |
|-------|---|----------|
| (i) | What is the velocity of the particle after 4 seconds? | 1 |
| (ii) | What is the particle's acceleration after 4 seconds? | 1 |
| (iii) | At what time after $t=0$ is the particle at rest? | 1 |
| (iv) | At what time does the particle change direction? | 1 |
| (v) | Explain what the shaded area represents. | 1 |

Question 15 continues on page 13

Question 15 (continued)

Marks

- (c) In Daniel's first year of fulltime employment his annual salary is \$75 000.
At the beginning of the second year his salary increases to \$79 000.
His salary continues to increase by \$4000 at the beginning of each successive year.
- (i) What will Daniel's salary be at the beginning of his 25th year?. **1**
- (ii) Calculate Daniel's total earnings after working for 25 years. **1**
- (iii) During which year of employment will his total earnings first exceed \$2 000 000? **2**
- (d) Determine the range of values of x when the curve $y = x^2 \ln x$ is concave up. **3**

End of Question 15

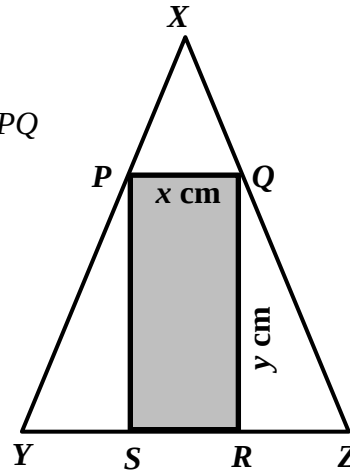
Question 16 (15 marks) Start a NEW booklet.**Marks**

- (a) Evaluate $\int_1^2 \frac{e^x}{e^x - 1} dx$ expressing the answer in simplified form.

3

- (b) In $\triangle XYZ$, $XY = XZ = 13$ cm and $YZ = 10$ cm.

A rectangle $PQRS$ is inscribed in the triangle with PQ parallel to YZ . Let $PQ = x$ cm and $QR = y$ cm.



- (i) Find the perpendicular height of $\triangle XYZ$.

1

- (ii) Using similar triangles, show that

$$y = 12 - \frac{6x}{5}.$$

2

- (iii) If the area of the rectangle $PQRS$ is A cm², find the maximum value of A .

2

- (c) Consider the function $f(x) = \frac{4}{x^2 + 1}$.

- (i) Show that $f(x) = \frac{4}{x^2 + 1}$ is an even function.

1

- (ii) Explain why there are no x -intercepts for $f(x) = \frac{4}{x^2 + 1}$.

1

- (iii) Find any stationary points and determine their nature.

3

- (iv) Neatly sketch the graph of $y = f(x)$.

2**End of paper**

Multiple Choice Section:

- 1.B 2.A 3.C 4.B 5.D
 6.D 7.B 8.C 9.A 10.D

Question 1.

$$2c = 1 - a$$

$$\therefore c = \frac{1 - a}{2} \quad \text{-----} \boxed{B}$$

Question 2.

$$f(1) = 2 - 3 + 4 = 3$$

$$f(-1) = 2 + 3 + 4 = 9$$

$$f(1) - f(-1) = -6 \quad \text{-----} \boxed{A}$$

Question 3.

$$\text{decreasing exponentially} \quad \text{-----} \boxed{C}$$

Question 4.

$$T_2 = -3 \rightarrow ar = -3 \text{-----} \boxed{1}$$

$$T_5 = 192 \rightarrow ar^4 = 192 \text{-----} \boxed{2}$$

$$\boxed{2} \div \boxed{1} \rightarrow r^3 = -64$$

$$\therefore r = -4 \quad \text{-----} \boxed{B}$$

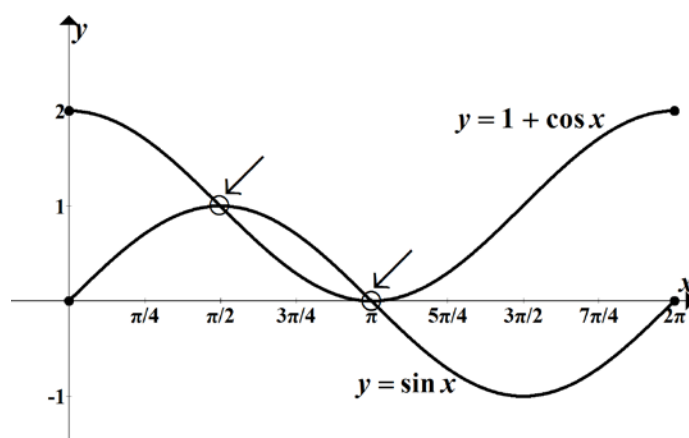
Question 5.

$$(x - 3)^2 + (y - 5)^2 = 4 \quad \text{-----} \boxed{D}$$

Question 6.

$$A = \frac{5-1}{6} (f(1) + 4f(3) + f(5))$$

$$= \frac{2}{3} (4a + b) \quad \text{-----} \boxed{D}$$

Question 7.

$$\therefore 2 \text{ points of intersection} \quad \text{-----} \boxed{B}$$

Question 8.

Using the test point (3,3) it can be determined that each inequality is true – hence III ----- $\boxed{C}$

Question 9.

$OA + OB$ represents the sum of the roots.

$$\therefore OA + OB = -\frac{b}{a} = -b \quad \text{-----} \boxed{A}$$

Question 10.

$$\therefore \log_{10} y - \log_{10} x = \log_{10} z - \log_{10} y$$

$$\therefore 2\log_{10} y = \log_{10} x + \log_{10} z$$

$$\therefore \log_{10} y^2 = \log_{10} (xz)$$

$$\therefore y^2 = xz \quad \text{-----} \boxed{D}$$

Question 11

a) $e^{-\frac{\pi+3}{2}} = 0.04638420$

$$\approx 0.0464 \text{ (3Sig fig)}$$

b) $50 - 2x^2 = 2(25 - x^2)$

$$= 2(5 - x)(5 + x)$$

c) $\frac{1}{x^2 - 1} - \frac{1}{x - 1} = \frac{1}{(x - 1)(x + 1)} - \frac{x + 1}{(x - 1)(x + 1)}$

$$= \frac{1 - x - 1}{(x - 1)(x + 1)} = \frac{-x}{(x - 1)(x + 1)}$$

d) $3x - 5 = 4$ or $3x - 5 = -4$

$$3x = 9 \text{ or } 3x = 1$$

$$\therefore x = 3 \text{ or } x = \frac{1}{3}$$

e) $LHS = \frac{1}{3 - \sqrt{2}} \times \frac{3 + \sqrt{2}}{3 - \sqrt{2}}$

$$= \frac{3 + \sqrt{2}}{7} \rightarrow a = \frac{3}{7}, b = \frac{1}{7}$$

f) Domain: all real $x \geq \frac{1}{2}$

Range: all real $y \geq 0$

g) $\int_0^{\frac{\pi}{2}} \sec^2 \frac{x}{3} dx = 3 \left[\tan \frac{x}{3} \right]_0^{\frac{\pi}{2}}$

$$= 3 \left(\tan \frac{\pi}{6} - \tan 0 \right) = \frac{3}{\sqrt{3}}$$

Question 12.

a) $\frac{d}{dx}(\sqrt{x^3 + 1}) = \frac{1}{2}(x^3 + 1)^{-\frac{1}{2}} \times 3x^2$

$$= \frac{3x^2}{2\sqrt{x^3 + 1}}$$

b) $81 - 27 + 9 - \dots \rightarrow a = 81, r = -\frac{1}{3}$

$$\therefore S_{\infty} = \frac{81}{1 - \left(-\frac{1}{3}\right)} = \frac{243}{4}$$

c) $\int (3x - 2)^5 dx = \frac{(3x - 2)^6}{18} + C$

d) $\sin \frac{3\pi}{4} + \cos \frac{7\pi}{6} = \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2}$

e) $\frac{dy}{dx} = 3x^2 - 5$

$$\therefore y = x^3 - 5x + C$$

when $x = 1, y = 1 \rightarrow C = 5$

$$\therefore y = x^3 - 5x + 5$$

f) $V = \frac{\pi t^3}{12} \text{ cm}^3 \rightarrow \frac{dV}{dt} = \frac{\pi t^2}{4}$

when $t = 1, \frac{dV}{dt} = \frac{\pi}{4} \text{ cm}^3/\text{s}$

g) i) $6y = (x^2 + 2x + 1) + 12$

$$\therefore 6(y - 2) = (x + 1)^2$$

hence the vertex is $(-1, 2)$

ii) for the focal length $4a = 6 \rightarrow a = \frac{3}{2}$

Question 13

a) $\frac{d}{dx}(e^{\cos x}) = -\sin x e^{\cos x}$

b) $d = \frac{|2 \times 1 - 5 \times -3 - 4|}{\sqrt{2^2 + (-5)^2}} = \frac{13}{\sqrt{29}}$

c) $\angle FCE = 60^\circ$ [$\triangle FCE$ is equilateral]

$\angle BCD = 90^\circ$ [$\triangle ABCD$ is a square]

$$\therefore \angle BCF = 360^\circ - 130^\circ - 90^\circ - 60^\circ = 80^\circ$$

and since $BC = CF$ as ($DC = CE$)

$$\text{then } \theta = \angle CBF = \frac{180^\circ - 80^\circ}{2} \quad [\angle\text{'s opp} = \text{sides}]$$

$$= 50^\circ \quad [\text{in isosceles } \triangle]$$

d) i) Product of roots = 1

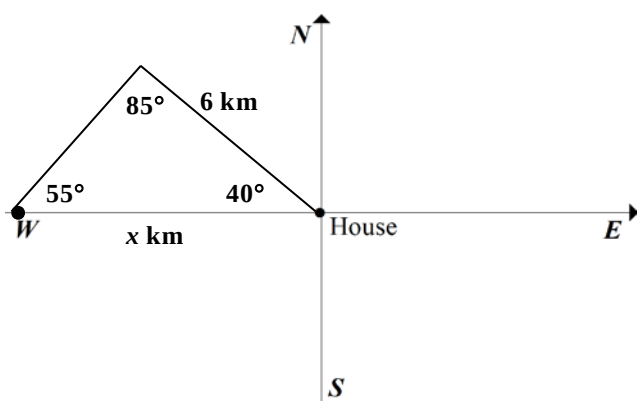
$$\therefore \frac{k}{3} = 1 \rightarrow k = 3$$

ii) $\Delta = 0$ for equal roots

$$\Delta = b^2 - 4ac = 100 - 12k$$

$$100 = 12k \rightarrow k = \frac{25}{3}$$

e)



$$\frac{x}{\sin 85^\circ} = \frac{6}{\sin 55^\circ} \rightarrow x = \frac{6 \sin 85^\circ}{\sin 55^\circ}$$

$$\therefore x = 7.3 \text{ km}$$

$$\text{f) } \int_0^3 -x^2 + ax + 12 \, dx = 45$$

$$\therefore \left[-\frac{x^3}{3} + \frac{ax^2}{2} + 12x \right]_0^3 = 45$$

$$-9 + \frac{9a}{2} + 36 = 45 \rightarrow a = 4$$

Question 14.

a) i) $A(-2, 0); B(0, 4);$

$$\text{ii) } m_{L_1} = 2 \rightarrow m_{L_2} = -\frac{1}{2}$$

$$\therefore L_2: y - 4 = -\frac{1}{2}(x - 0) \text{ using } B(0, 4)$$

$$\therefore 2y - 8 = -x$$

$$\therefore x + 2y - 8 = 0$$

iii) $y = 2x$

iv) Solve $y = 2x$ and $x + 2y - 8 = 0$

$$\therefore x + 2(2x) - 8 = 0$$

$$5x = 8 \rightarrow x = \frac{8}{5}$$

$$\text{when } x = \frac{8}{5}, y = 2 \times \frac{8}{5} = \frac{16}{5}$$

v) C has coordinates $(8, 0)$.

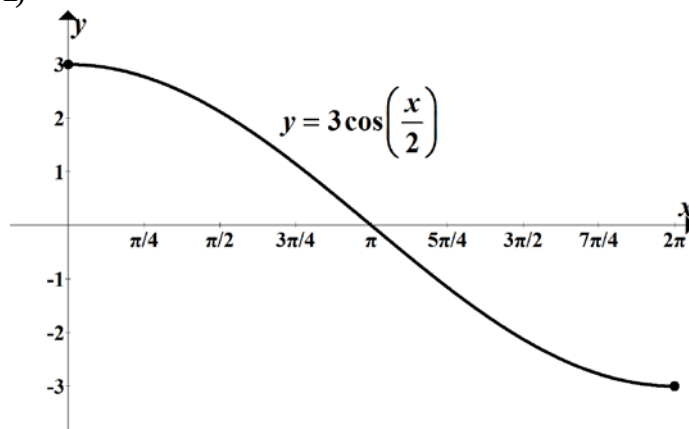
Area of quadrilateral $OABD$ =

Area of $\triangle ABC$ - Area of $\triangle ODC$

$$= \frac{1}{2} \times 10 \times 4 - \frac{1}{2} \times 8 \times \frac{16}{5} = \frac{36}{5} u^2$$

$$\text{b) i) } P = \frac{2\pi}{\frac{1}{2}} = 4\pi$$

ii)



$$\begin{aligned}
\text{c) } LHS &= \frac{\sin \theta}{\cos \theta} \times \frac{1}{\sin \theta} - \cos \theta \\
&= \frac{1}{\cos \theta} - \cos \theta \\
&= \frac{1 - \cos^2 \theta}{\cos \theta} \\
&= \frac{\sin^2 \theta}{\cos \theta} \\
&= \frac{\sin \theta}{\cos \theta} \times \frac{\sin \theta}{1} \\
&= \sin \theta \tan \theta = RHS
\end{aligned}$$

$$\text{d) } (x - 5)(x + 2) = 0$$

$$\therefore x = 5 \text{ or } x = -2$$

but since $x > 0$ then $x = 5$ only

Question 15.

$$\text{a) } V = \pi \int_1^8 x^2 dy \quad \left[y = x^3 \rightarrow x^2 = y^{\frac{2}{3}} \right]$$

$$\therefore V = \pi \int_1^8 y^{\frac{2}{3}} dy$$

$$= \pi \left[\frac{3}{5} y^{\frac{5}{3}} \right]_1^8$$

$$= \frac{3\pi}{5} \left(8^{\frac{5}{3}} - 1^{\frac{5}{3}} \right)$$

$$= \frac{3\pi}{5} (31) = \frac{93\pi}{5} u^3$$

$$\text{b) i) } 12 \text{ m/s}$$

$$\text{ii) } 0 \text{ m/s}^2$$

$$\text{iii) } t = 9 \text{ s}$$

$$\text{iv) } t = 9 \text{ s}$$

v) The distance travelled in the first 9 seconds.

$$\text{c) i) } \$75000 + 24 \times 4000 = \$171000$$

ii) Total earnings for 25 years =

$$\$75000 + \$79000 + \$83000 + \dots + \$171000$$

$$= \frac{25}{2} (\$75000 + \$171000)$$

$$= \$3075000$$

$$\text{iii) } 2000000 = \frac{n}{2} (2 \times 75000 + (n - 1) \times 4000)$$

$$4000000 = 150000n + 4000n^2 - 4000n$$

$$\therefore 2n^2 + 73n - 2000 = 0$$

$$n = \frac{-73 \pm \sqrt{73^2 - 4 \times 2 \times -2000}}{4}$$

and since $n > 0$, $n \approx 18.26$

hence during the 19th year.

d) Concave up when $y'' > 0$

$$\therefore y' = x^2 \times \frac{1}{x} + 2x \times \ln x$$

$$= x + 2x \ln x$$

$$\therefore y'' = 1 + 2x \times \frac{1}{x} + 2 \times \ln x$$

$$= 3 + 2 \ln x$$

$$\therefore 3 + 2 \ln x > 0$$

$$\therefore \ln x > -\frac{3}{2} \rightarrow x > e^{-3/2}$$

Question 16.

$$a) \int_1^2 \frac{e^x}{e^x - 1} dx = \left[\ln(e^x - 1) \right]_1^2$$

$$= \ln(e^2 - 1) - \ln(e - 1)$$

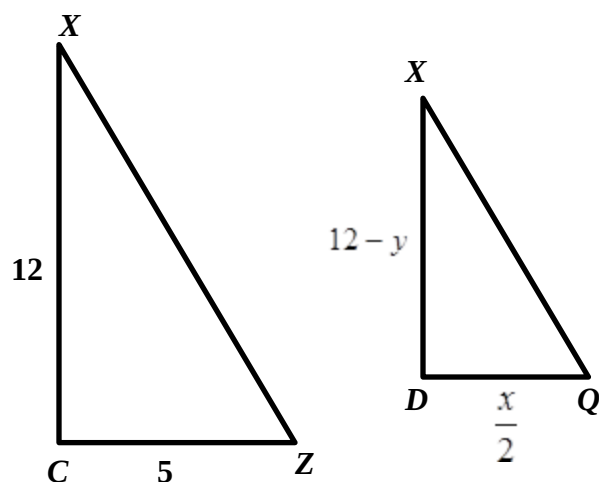
$$= \ln \frac{(e-1)(e+1)}{(e-1)}$$

$$= \ln(e+1)$$

b) i) Perpendicular height is 12 cm (Pythagoras)

ii) Let D be the midpoint of PQ .

$$\therefore DQ = \frac{x}{2} \text{ and } XD = 12 - y.$$



The triangles are similar (equiangular), hence corresponding sides are in the same ratio.

$$\therefore \frac{12-y}{\frac{x}{2}} = \frac{12}{5}$$

$$\therefore 6x = 60 - 5y$$

$$\therefore y = 12 - \frac{6x}{5}$$

$$iii) A = xy \rightarrow A = x \left(12 - \frac{6x}{5} \right)$$

$$\therefore A = 12x - \frac{6x^2}{5}$$

For a maximum area $\frac{dA}{dx} = 0$ and $\frac{d^2A}{dx^2} < 0$.

$$\frac{dA}{dx} = 12 - \frac{12x}{5}; \frac{d^2A}{dx^2} = -\frac{12}{5} < 0 \text{ hence maximum.}$$

$$\therefore 12 - \frac{12x}{5} = 0 \rightarrow x = 5 \text{ cm}$$

$$\text{Hence maximum area} = 5 \times \left(12 - \frac{6 \times 5}{5} \right) = 30 \text{ cm}^2$$

c) i) A function is even if $f(x) = f(-x)$.

$$f(-x) = \frac{4}{(-x)^2 + 1}$$

$$= \frac{4}{x^2 + 1} = f(x), \text{ hence even.}$$

ii) x -intercepts occur when $f(x) = 0$, and since the numerator is constant then $f(x) \neq 0$.

$$iii) f(x) = 4(x^2 + 1)^{-1}$$

$$\therefore f'(x) = -8x(x^2 + 1)^{-2} = \frac{-8x}{(x^2 + 1)^2}$$

For stationary points $f'(x) = 0 \rightarrow x = 0$.

When $x = 0, y = 4$.

Test for maximum or minimum:

x	-1	0	1
$f'(x)$	2	0	-2

/ — \

Hence $(0, 4)$ is a maximum stationary point.

iv)

